



Introduction to Linux for Biological Sciences

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Outline

- Why Linux?
- Access of Linux
- Linux Fundamentals
 - File system navigation
 - File operations
 - Permissions
 - Help Resources
- Text Processing Tools
 - Viewing files
 - Power filters
 - Redirection
- Vim Text Editor

Experiments

- Install genomics analysis software (ANNOVAR) based on Linux systems
- View and understand standard file formats
- Use a bash script to calculate the number of RNA sequences on a specific chromosome

01. Introduction

Features of the Linux System

- Free-to-use & freely distributed
- Multi-user, multi-tasking, multi-threading and multi-CPU
- Security
 - Strict permission control and multi-level security mechanisms
 - Prevents malicious attacks and system crashes
- Stability
 - Runs for long periods without crashing
 - Meeting enterprise-level application demands
- Design Philosophy
 - Each command or application is small and performs only a single function.
 - Multiple programs can be linked together to complete complex tasks.

Linux vs. Windows

Feature	Windows	Linux
Interface	Primarily GUI-based , with a consistent graphical interface across versions.	Mainly command-line interface ; graphical interface varies by distribution.
Security	OS, applications, and interface are tightly integrated, posing security risks .	Modular design; the Linux kernel is independent of applications and the interface, preventing security vulnerabilities or crashes from affecting the entire system .
System Configuration	Configuration is stored in an encrypted database.	All configuration data is stored in plain text files, making it accessible without special interfaces and facilitating configuration transfers .
Learning	Complex system structure with frequent updates; knowledge and skills become obsolete quickly, making deep learning challenging.	Simple and stable system structure; knowledge and skills have long-term relevance, making deep learning relatively easier.
Software	Most functionalities require commercial software, which requires purchasing licenses.	Most software is freely available, though fewer options exist for the same functionality.

Why Linux for biologists?

- Most tasks in bioinformatics are more suited, or even exclusive to the command line
 - big data manipulation (>10GB per file)
 - model training and inference
 - massive parallelism
- Think of the command line as another tool you can add to your belt.
- As always, pick the best tool for the job.

Linux Distributions

- Red Hat Linux
 - Most widely used and recognized.
 - CentOS: Suitable for servers.
- Ubuntu Linux
 - User-friendly interface, easy to get started with, supports various hardware, and all distributions are free.
- SuSE Linux
 - Facilitates interaction with Windows, excellent hardware detection, user-friendly installation process, and graphical management tools.
- Gentoo Linux
 - Highly customizable
 - Very complex installation process, but best performance and ease of management once installed.

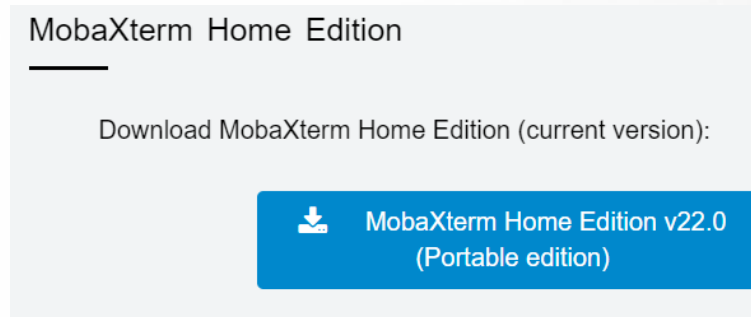


Getting Linux

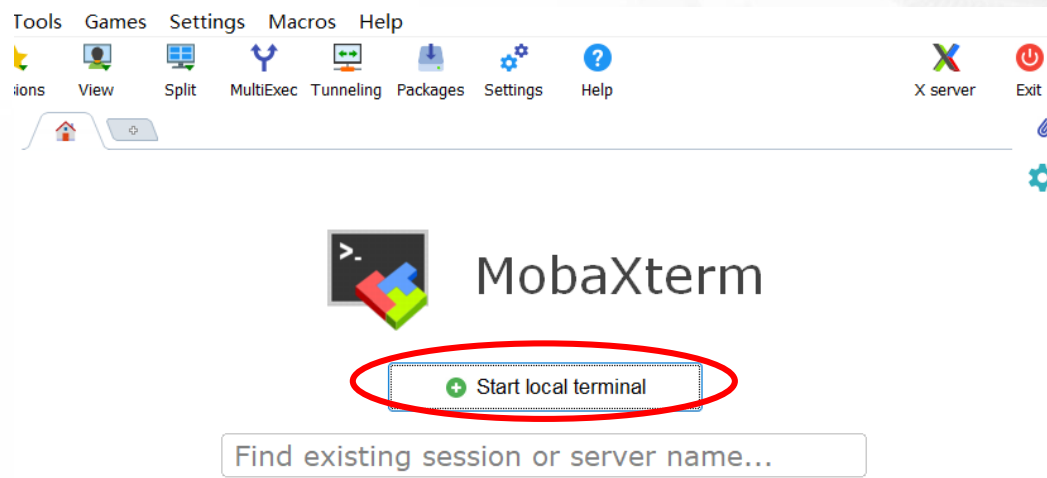
- Install using a boot disk
- Windows
 - Dual-boot system
 - Virtual machines
 - Windows Subsystem for Linux (WSL)
 - MobaXterm, cygwin
- MacOS
 - Virtual Machine
 - terminal

Installing MobaXterm

- Download: <https://mobaxterm.mobatek.net/download.html>



- Save to **C drive** → Extract to **MobaXterm** folder → Create a desktop shortcut



02. Linux Fundamentals

Shell

- Shell: a command-line interpreter that acts as an interface between users and the Linux kernel.
- Bash is the most commonly used shell and the default in most Linux distributions.

- Command Format:

operation

method

operand

- command [flags] [argument1] [argument2] ...
 - Commands, options, and parameters must be separated by spaces
 - Example: `ls`; `ls -lr`; `ls -lr /`
 - Order doesn't matter: `ls -lR` = `ls -R -l`
- History command:
 - `history [n]`
- Auto-completion: `<tab>`
 - for commands, directories, and files
- Copy/Paste: Select → Right-click

Linux File System Structure

- Everything in Linux is stored as a file
- Root directory (/): entry point → **Absolute path** (The GPS Coordinate)
 - /home/mobaxterm/Desktop
 - /mnt/d/
 - Unambiguous—works from **any location** in the system
- Current directory (.), Parent directory (..) → **Relative path** (The Local Landmark)
 - Desktop = ./Desktop
 - ../
 - Faster when working within a single project folder.
- Home Directory ~ Your 'lab bench' space

File System navigation

Command	Function	Biological Analogy	Parameter
pwd	print working directory	GPS in cell	
cd	change directory	Move microscope objective	
ls	list	Fluorescent tagging	-l
mkdir	make directory	Create new cell organelle	-p

Common Error:

Error Message	Solutions
No such file or directory	Check spelling with ls first Use Tab autocomplete to avoid typos
Permission denied	Use sudo (with instructor approval)
Is a directory	Use cd instead of cat

Practice

Command	Function	Biological Analogy	Parameter
pwd	print working directory	GPS in cell	
cd	change directory	Move microscope objective	
ls	list	Fluorescent tagging	-l
mkdir	make directory	Create new cell organelle	-p

- Practice (1 point)
 - Start at home: `cd ~`
 - Create a course folder: `mkdir [yourpath]/bioinfo`
 - Enter course directory using absolute path or relative path
 - Verify: `pwd`

File Operations

Command	Function	Syntax	Parameter
cp	copy	cp [options] source destination	-r(recursive)
mv	move	mv [options] source destination	
rm	remove	rm [options] file_or_directory	-rf (dangerous!)

Pathname Expansion

- ? Matches exactly one arbitrary character
- * Matches zero or more arbitrary characters

Common Error:

Error Message	Solutions
cp: omitting directory	Add -r for directories
rm: cannot remove	Check permissions with ls -l

rm Command

- Delete one or multiple files or directories within a directory.
 - -r (recursive): If omitted, rm will not delete directories
 - -f (force): Force deletion
 - -i (interactive): Prompt before deletion
- **WARNING! There is no recycle bin!** Always use ls first to confirm.
- **DO NOT execute rm -rf / under system privileges!**

Practice

Command	Function	Syntax	Parameter
cp	copy	cp [options] source destination	-r(recursive)
mv	move	mv [options] source destination	
rm	remove	rm [options] file_or_directory	-rf (dangerous!)

- Download the **ANNOVAR** package from official site to D:/
 - <https://annovar.openbioinformatics.org/en/latest/user-guide/download/>
- Copy the package to the course folder
 - /mnt/d or D:/
- verify using ls (0.5 point)
- Use bash to delete the package from D:/ (0.5 point)

Permissions

Use `ls -l [filename]` to check permissions.

For example: `-rw-r--r-- 1 root root 296K 11-26 10:03 bio.log`

There are two usage methods:

- Symbolic mode using letters and operators
- Numeric mode using numbers

The access permissions of a file or directory are divided into three groups, each represented by three bits:

- Read, write, and execute permissions for the file owner
- Read, write, and execute permissions for users in the same group as the owner
- Read, write, and execute permissions for other users in the system



Permission operations

chmod: change the access permissions of files or directories in Linux

Permission scope:

- **u** The current user of the file or directory
- **g** The current group of the file or directory
- **o** Users or groups other than the current user or group of the file or directory
- **a** All users and groups

Permission codes:

- **r** Read permission, represented by the number 4
- **w** Write permission, represented by the number 2
- **x** Execute permission, represented by the number 1

Common options:

- **-c** Report processing information when changes occur
- **-R** Apply changes to the specified directory and all its subdirectories

Examples:

- Grant all users execute permission for the file anno.sh
 - **chmod a+x t.anno.sh**
- Revoke all existing permissions, then grant the owner read permission and output processing information
 - **chmod u=r t.log -c**
- Assign read, write, and execute permissions (4+2+1) to the file owner, read and execute permissions to the group, and execute permission to other users
 - **chmod 751 t.log -c**
 - **chmod u=rwx,g=rx,o=x t.log -c**
- Lock down sensitivity data
 - **chmod 600 patient.vcf**

Permission Operations

- sudo
 - The Linux sudo command executes instructions with system administrator privileges, meaning commands run via `sudo` are executed as if they were performed by the root user.
 - Usage permission: Only users listed in `/etc/sudoers` can use `sudo`.
 - Example: Execute the previous command with root privileges
 - `sudo !!`

Viewing Help

- **[command] --help**

- a brief description of the command and a list of options.
- can be used for most commands but not all.
- options or parameters enclosed in [] are optional.
- a|b|c means only one of a, b, or c can be used.
- e.g. cp --help

- man command

- each command has a corresponding manual page.
- use arrow keys to move up, down, left, or right.
- press q to exit.

- Bing.com

- LLMs

Practice (Run and explain the code, write the source of help, 1 point):

```
tar -zxvf annovar.latest.tar.gz
```

03. Text Processing tools

Viewing Files

- **cat (Concatenate Files)**

- Display, combine, or create files.
- View small files (e.g., gene lists, metadata).
- Merge multiple sequence files (FASTA/FASTQ) into one.
- Ctrl + C to interrupt

Practice:

```
cat  
[your_path]/bioinfo/annovar/example/R  
EADME
```

- **less (Scrollable File Viewer)**

- View large files interactively (scroll/search without loading the entire file).
- -S or --chop-long-lines
- -N or --LINE-NUMBERS
- Press `/` to search
 - e.g., `/chr22` to find chromosome 22
- Press `q` to exit

Practice:

```
less  
[your_path]/bioinfo/annovar/humandb/hg3  
8_refGeneWithVerMrna.fa and search for  
chr22 (1 point)
```

View the expression data in class2

- **head & tail (View File Start/End)**

- Quickly check file structure (e.g., headers in CSV/TSV files)
- Monitor ongoing processes
- Why would you use tail -f during an ongoing experiment?

Power filters

- **grep**
 - Purpose: Search for patterns in text using regular expressions.
 - Key Flags:
 - -i (case-insensitive)
 - -v (invert match, exclude lines)
 - -c (count matches)
 - Processes line by line, suitable for large texts.
 - Biological Use Cases:
 - Extract gene/protein sequences from FASTA files.
 - Filter logs for specific experiment IDs or errors.
- **sed**
- **awk**

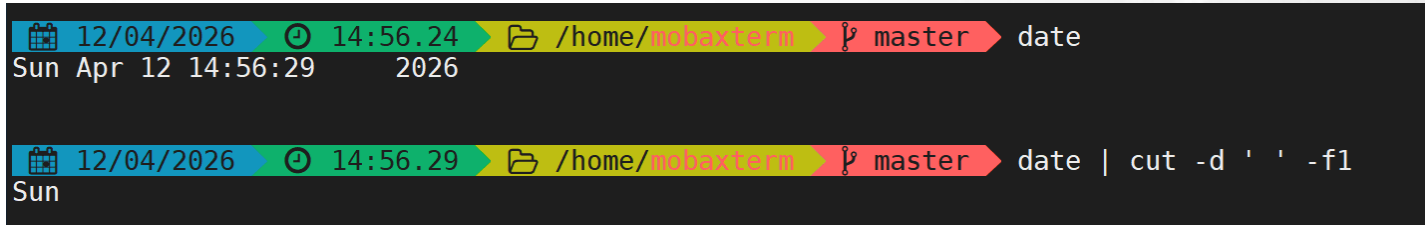
Text Analysis with `wc`

- `wc [OPTIONS] [FILE...]`
- Core Functions: It counts lines, words, and bytes/characters.

Option	What it counts	Bioinformatic Application Example
(none)	Lines, Words, Bytes	General file overview.
-l	Lines	Crucial: Count FASTQ reads (lines/4), FASTA sequences (count '>'), file entries.
-w	Words (space-separated)	Less common, but useful for columnar data.
-c	Bytes	File size in bytes.
-m	Characters	File size in characters (differs for multi-byte encodings).

Pipelines

- Use the output of one command as the input for another.
- `command1 | command2 | command3 | ...`
- Example:
 - `date | cut -d ' ' -f1`



```
12/04/2026 14:56.24 /home/subashish master date
Sun Apr 12 14:56:29 2026

12/04/2026 14:56.29 /home/subashish master date | cut -d ' ' -f1
Sun
```

- `grep "chrX" hg38_refGeneWithVerMrna.fa | less`

Input/Output Redirection

- Default Input/Output Devices:
 - Standard Output: STDOUT, 1
 - Standard Input: STDIN, 0
 - Standard Error Output: STDERR, 2
- Redirection Operators:
 - Standard Input (< or <<)
 - Standard Output (> or >>)
 - Standard Error Output (2> or 2>>)
 - Example: `ls / /varr >ans.out 2> err.out`

Practice (1 point):

- In a FASTA file `[file_path]/bioinfo/annovar/humandb/hg38_refGeneWithVerMrna.fa`, filter lines containing 'chr21'.
- Output them to `hg38_chr21.txt`.
- View the new file.

FASTA

A well-structured data format can efficiently process data, reduce storage space, and increase computational speed. For the massive data generated by deep sequencing, properly and effectively organizing data storage is crucial for accelerating downstream analysis.

- representing either nucleotide sequences or peptide sequences
 - begins with a single-line description, followed by lines of sequence data.
 - the description line: a greater-than (">") symbol in the first column
 - blank lines are not allowed in the middle of FASTA input
- For genome browsing, pipeline input etc.

A	alanine	P	proline
B	aspartate/asparagine	Q	glutamine
C	cystine	R	arginine
D	aspartate	S	serine
E	glutamate	T	threonine
F	phenylalanine	U	<i>selenocysteine</i>
G	glycine	V	valine
H	histidine	W	tryptophan
I	isoleucine	Y	tyrosine
K	lysine	Z	glutamate/glutamine
L	leucine	X	any
M	methionine	*	translation stop
N	asparagine	-	<i>gap of indeterminate length</i>

```
>gi|129295|sp|P01013|OVAX_CHICK GENE X PROTEIN (OVALBUMIN-RELATED)
QIKDLLVSSSTDLDTTLLVLVNAIYFKGMWKTAFAEDTREMPFHVTQESKPVQMMCMNNSFNVATLPAE
KMKILELPFASGDLSMLVLLPDEVSDLERIEKTINFEKLTEWTNPNTMEKRRVKVYLPQMKIEEKYNLTS
VLMALGMTDLFIPSANLTGISSAESLKISQAVHGAFMELSEDGIEMAGSTGVIEDIKHSPSEQFRADHP
FLFLIKHNPTNTIVYFGRYWSP
```

04. Vim

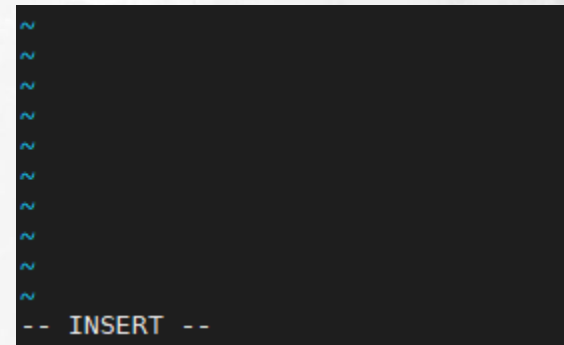


Why Vim?

- Lightweight, preinstalled on most Linux systems.
- Efficient for editing large files (e.g., genome sequences, logs).
- Keyboard-driven workflow minimizes reliance on GUIs.

Basic Vim Operations

- Create or Open a File: `vim <FILE_NAME>`
- Exit without saving: `q!`
- Modify a file:
 - Press `Esc` to ensure you are in Normal mode.
 - Press `i` to enter Insert mode.
 - Make your modifications.
 - Press `Esc` to return to Normal mode.
 - Type: `wq` and press `Enter` to save and exit.



Modes

- Normal mode
 - Navigate, delete, copy/paste
 - Press Esc to switch
- Insert mode
 - Type/edit text (press i to enter).
- Command Mode
 - Save, quit, or search (press : to enter).
- Vim also has other modes, such as Visual mode, Select mode, and Command mode.

Navigation

- `^` → Jump to line start.
- `$` → Jump to line end.
- `nG` → Jump to line n.
- `gg` → Jump to file start.
- `G` → Jump to file end.
- `h, j, k, l` → Move left/down/up/right.

Searching

- Press **Esc** to enter Normal mode.
- Type **“.”**
- Type **/<SEARCH_KEYWORD>** to search.
- Matches will be highlighted.
- Press **n** to find the next match and **N** for the previous one.

Editing

- Insert Mode
- Normal Mode :
 - Press **v** and move the cursor to select text.
 - **y** → Copy
 - **yy** → Copy (yank) line.
 - **p** → Paste copied/yanked text.
 - **dd** → Delete entire line.
 - **u** → Undo last change.
 - **Ctrl+R** redo

Saving/Exiting (Command Mode)

- Command Mode:
 - :w → Save.
 - :q! → Quit without saving.
 - :wq → Save and quit.

Bash Scripting

- Any command entered in the Linux terminal prompt can be placed in a Bash script file.
- **Bash Script File Format**
- **Variables**
 - In Bash, all variables are strings.
 - `VAR=10` # No spaces around "="
 - `echo $VAR`
- Lines starting with `#` are comments.

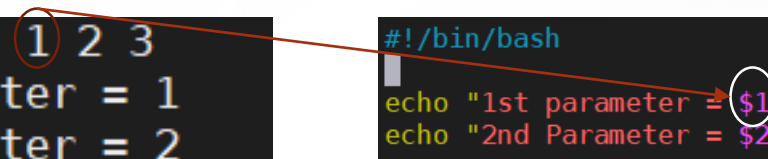
```
1 #!/bin/bash
2
3 function1()
4 {
5 }
6
7 function2()
8 {
9 }
10
11 # 主程序
12 echo "hello world" # 打印hello world
13 ...
14 ...
15 ...
```

Running a Shell Script

- `chmod +x test.sh` # Make the script executable
- `./test.sh` # Run the script
- `/bin/sh test.sh` # Execute using /bin/sh
- Passing arguments:
 - `$n`
 - `./test.sh 1 2 3` # Run script with three arguments

```
$. /test.sh 1 2 3  
1st parameter = 1  
2nd Parameter = 2
```

```
#!/bin/bash  
echo "1st parameter = $1 "  
echo "2nd Parameter = $2 "  
~
```

A red arrow originates from the circled number '1' in the command line of the first terminal window and points to the '\$1' variable in the script of the second terminal window. Another red arrow originates from the circled number '2' in the command line of the first terminal window and points to the '\$2' variable in the script of the second terminal window.

Practice (Write a Bash script, 1 point):

The script accepts a single FASTA-formatted file as input and outputs the **number of sequences on chr21**.

Conditional Statements

- Conditional statements allow your scripts to:
 - **Check prerequisites** (e.g., "If the FASTQ file is missing, exit with an error").
 - **Handle errors gracefully** (e.g., "If the alignment tool fails, send an alert").
 - **Run different commands** based on different conditions (e.g., Process RNA vs. DNA files differently).
- Syntax
 - if [arg1 operator arg2] ; then command
 - if [arg1 operator arg2] ; then command1; else command2; fi
 - **Spaces cannot be omitted!**
 - **Control flow cannot be empty!**
 - **Quoting variables** to prevents errors if variables contain spaces.
- Operators
 - File Operators
 - e FILE: True if FILE exists (regardless of type).
 - f FILE: True if FILE is a regular file (not a directory).
 - d DIRECTORY: True if DIRECTORY exists.
 - Numeric Operators
 - eq → Equal to
 - ne → Not equal to
 - gt → Greater than
 - lt → Less than
 - String Operators
 - == → Equal
 - != → Not equal
 - ! → Logical NOT

```
#!/bin/bash
score=$1
if [ $score -gt 60 ];then
    printf "Your score is ${score}.\nPass!\n"
else
    printf "Your score is ${score}.\nFail!\n"
fi
```

Practice (1 point):

Based on the previous script,
check if the input file exists.

Loop Statements

- Loops allow you to:
 - Automate repetitive tasks
 - Scale scripts effortlessly
 - Reduce human error
- Syntax: **for** Loops

```
for var in item1 item2 ... itemN
do
    command1
    command2
    ...
    commandN
done
```

\$var

```
while condition
do
    command
done
```

```
until condition
do
    command
done
```

Practice (1 points):

- Based on the previous script, compare the number of sequences on all autosome
hint: {1..n} printf

```
[student@studentvm1 ~]$ X=0 ; while [ true ] ; do echo $((X++)) ; done | head
0
1
2
3
4
[student@studentvm1 ~]$ X=0 ; until false ; do echo $((X++)) ; done | head
5
0
6 1
7 2
8 3
9 4
5
6
7
8
9
```


Practice

- Is the number of RNA sequences on each chromosome proportional to the length of the chromosome? Visualize the results using R (1 point)
 - <https://www.ncbi.nlm.nih.gov/grc/human/data>
- Why or why not? Discuss the factors that might influence the relationship (1 point)
 - Hint: gene density

Human Genome Assembly GRCh38.p14
Information on tiling path files (TPFs) for the assembly is available at [TPF Overview](#).

Other assembly versions:

Chromosome lengths are calculated by summing the length of the placed scaffolds and estimated gaps.

Chromosome	Total length (bp)	GenBank accession	RefSeq accession
1	248,956,422	CM000663.2	NC_000001.11
2	242,193,529	CM000664.2	NC_000002.12
3	198,295,559	CM000665.2	NC_000003.12
4	190,214,555	CM000666.2	NC_000004.12
5	181,538,259	CM000667.2	NC_000005.10
6	170,805,979	CM000668.2	NC_000006.12
7	159,345,973	CM000669.2	NC_000007.14
8	145,138,636	CM000670.2	NC_000008.11
9	138,394,717	CM000671.2	NC_000009.12
10	133,797,422	CM000672.2	NC_000010.11
11	135,086,622	CM000673.2	NC_000011.10
12	133,275,309	CM000674.2	NC_000012.12
13	114,364,328	CM000675.2	NC_000013.11
14	107,043,718	CM000676.2	NC_000014.9
15	101,991,189	CM000677.2	NC_000015.10
16	90,338,345	CM000678.2	NC_000016.10
17	83,257,441	CM000679.2	NC_000017.11
18	80,373,285	CM000680.2	NC_000018.10
19	58,617,616	CM000681.2	NC_000019.10
20	64,444,167	CM000682.2	NC_000020.11
21	46,709,983	CM000683.2	NC_000021.9
22	50,818,468	CM000684.2	NC_000022.11
X	156,040,895	CM000685.2	NC_000023.11
Y	57,227,415	CM000686.2	NC_000024.10

Experiment

- Install genomics analysis software ANNOVAR on a Linux system.
- Explore the relationship between the number of RNA sequences on chromosomes and the length of DNA.
 - Understand standard file formats and calculate the number of sequences on a specific chromosome in a FASTA file.
 - Compare the number of sequences on all autosome.
 - Is the number of RNA sequences on each chromosome proportional to the length of the chromosome?
 - Think: Discuss the factors that might influence the relationship.
- Save the code and results in a Word or PDF file (a complete lab report is not required). Name the file as **“StudentID_Name_ExpX_ClassY”**.
- If AI assistance is used to complete the experiment or report, the purpose and prompts **must be** explicitly stated; otherwise, the assignment will be graded as 0 points.

Learn More

- <https://www.runoob.com/linux/linux-shell.html>
- <https://www.tutorialspoint.com/vim/index.htm>
- 公众号： 康康生信

Thank you!